Vector-Valued Functions

ANSWER KEY

Section 1: Position, Velocity, and Acceleration Vectors

1 A particle has position vector $\mathbf{r}(t) = \langle t^3 - 2t, \sin(2t) \rangle$. Find the acceleration vector $\mathbf{a}(t)$ at $t = 0$.

a. $\langle 0, 0 \rangle$ ✓

b. $\langle -2, 2 \rangle$

c. $\langle 0, -4 \rangle$

d. $\langle 6, 0 \rangle$

$\mathbf{v}(t) = \langle 3t^2 - 2, 2\cos(2t) \rangle$. $\mathbf{a}(t) = \langle 6t, -4\sin(2t) \rangle$. At $t = 0$: $\mathbf{a}(0) = \langle 0, 0 \rangle$.

2 If the velocity vector of a particle is $\mathbf{v}(t) = \langle 3t^2, 4t \rangle$ and $\mathbf{r}(0) = \langle 1, 5 \rangle$, find $\mathbf{r}(1)$.

a. $\langle 2, 7 \rangle$ ✓

b. $\langle 1, 2 \rangle$

c. $\langle 4, 9 \rangle$

d. $\langle 3, 4 \rangle$

$\mathbf{r}(t) = \langle t^3 + 1, 2t^2 + 5 \rangle$. At $t = 1$: $\mathbf{r}(1) = \langle 1 + 1, 2 + 5 \rangle = \langle 2, 7 \rangle$.

3 What is the magnitude of the acceleration vector $\mathbf{a}(t) = \langle 3, -4 \rangle$?

a. 5 ✓

b. 7

c. $\sqrt{7}$

d. 1

$\|\mathbf{a}\| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = 5$.

Section 2: Finding Position from Velocity & Acceleration

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4 Given $\mathbf{a}(t) = \langle 0, -32 \rangle$ with initial velocity $\mathbf{v}(0) = \langle 20, 64 \rangle$, what is $\mathbf{v}(2)$?

a. $\langle 20, 0 \rangle$ ✓

b. $\langle 20, -32 \rangle$

c. $\langle 0, 0 \rangle$

d. $\langle 20, 32 \rangle$

$\mathbf{v}(t) = \langle 20, -32t + 64 \rangle$. At $t = 2$: $\mathbf{v}(2) = \langle 20, -64 + 64 \rangle = \langle 20, 0 \rangle$.

5 In the trajectory above, what is the maximum height achieved if initial position is $\mathbf{r}(0) = \langle 0, 0 \rangle$?

a. 64 **feet** ✓

b. 32 feet

c. 128 feet

d. 16 feet

Maximum height occurs when $v_y(t) = 0 \Rightarrow t = 2$. $y(t) = -16t^2 + 64t$. At $t = 2$: $y(2) = -16(4) + 64(2) = -64 + 128 = 64$ feet.

6 If a particle has $\mathbf{v}(t) = \langle \frac{1}{t+1}, 2e^{2t} \rangle$ and $\mathbf{r}(0) = \langle 3, 4 \rangle$, find the x-component $x(t)$.

a. $x(t) = \ln(t+1) + 3$ ✓

b. $x(t) = \ln(t+1)$

c. $x(t) = -\frac{1}{(t+1)^2} + 3$

d. $x(t) = e^{t+1} + 2$

$x(t) = \int \frac{1}{t+1} dt = \ln(t+1) + C$. Since $x(0) = 3$, $\ln(1) + C = 3 \Rightarrow C = 3$.

Section 3: Speed, Direction, and Displacement

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7 What is the direction angle θ of motion when $\mathbf{v}(t) = \langle -2, 2\sqrt{3} \rangle$?

a. $\frac{2\pi}{3}$ ✓

b. $\frac{\pi}{3}$

c. $\frac{5\pi}{6}$

d. $\frac{4\pi}{3}$

$\tan\theta = \frac{v_y}{v_x} = \frac{2\sqrt{3}}{-2} = -\sqrt{3}$. In Quadrant II, $\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$.

8 Which vector represents the net displacement of a particle from $t = 1$ to $t = 4$?

a. $\int_1^4 \mathbf{v}(t)dt$ ✓

b. $\int_1^4 \|\mathbf{v}(t)\|dt$

c. $\mathbf{v}(4) - \mathbf{v}(1)$

d. $\mathbf{a}(4) - \mathbf{a}(1)$

Displacement vector is $\Delta\mathbf{r} = \mathbf{r}(4) - \mathbf{r}(1) = \int_1^4 \mathbf{v}(t)dt$.

9 If speed is constant, what must be true about the acceleration vector and the velocity vector?

a. $\mathbf{a}(t) \cdot \mathbf{v}(t) = 0$ **(orthogonal)** ✓

b. $\mathbf{a}(t)$ is zero

c. $\mathbf{a}(t)$ is parallel to $\mathbf{v}(t)$

d. $\mathbf{a}(t) = -\mathbf{v}(t)$

Since speed squared $\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$ is constant, differentiating yields $2\mathbf{v} \cdot \mathbf{a} = 0$, meaning acceleration is always perpendicular to velocity.

Section 4: Distance Traveled with Vector-Valued Functions

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10 Write an integral for the total distance traveled by a particle from $t = 0$ to $t = 5$ with $\mathbf{v}(t) = \langle \cos(t^2), e^{\sqrt{t}} \rangle$.

a. $\int_0^5 \sqrt{\cos^2(t^2) + e^{2\sqrt{t}}}\, dt$ ✓

b. $\int_0^5 (\cos(t^2) + e^{\sqrt{t}}) dt$

c. $\sqrt{\int_0^5 \cos^2(t^2) dt + \int_0^5 e^{2\sqrt{t}} dt}$

d. $\int_0^5 \|\mathbf{a}(t)\| dt$

Distance traveled is $\int_0^5 \|\mathbf{v}(t)\| dt = \int_0^5 \sqrt{(\cos(t^2))^2 + (e^{\sqrt{t}})^2}\, dt = \int_0^5 \sqrt{\cos^2(t^2) + e^{2\sqrt{t}}}\, dt$.

11 A particle moves with speed $\sqrt{9 + 16t^2}$. What is the total distance traveled from $t = 0$ to $t = 1$?

a. $\int_0^1 \sqrt{9 + 16t^2}\, dt$ ✓

b. $\sqrt{25} = 5$

c. $\frac{25}{2}$

d. 4

Distance is the definite integral of speed over the time interval: $\int_0^1 \sqrt{9 + 16t^2}\, dt$.

12 If a particle starts at $(2, 3)$ and has displacement $\langle -4, 7 \rangle$, what is its final position?

a. $(-2, 10)$ ✓

b. $(6, -4)$

c. $(2, 10)$

d. $(-6, 4)$

Final position $= \langle 2 + (-4), 3 + 7 \rangle = \langle -2, 10 \rangle$.

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